

Orthographic-semantic consistency as a measure of graded morphological information

Patience R. Stevens^{1,3}, David C. Plaut^{2,3}

¹ Program for Interdisciplinary Education Research ² Neuroscience Institute ³ Department of Psychology
Carnegie Mellon University

Orthographic-semantic consistency

- Orthographic-semantic consistency (OSC): measure of how semantically similar stems are to other words containing them (Marelli and Amenta, 2018)
- Predicts lexical decision latencies; interacts with prime frequency to predict priming magnitudes (Amenta et al., 2020)
- Limitations of original OSC metric:
 1. Limited to measuring consistency of morphemes that exist as words
 2. Assumes semantic dominance of stand-alone stem

Cluster-centered variation



Original stem-centered calculation: Frequency-weighted mean of cosine similarity between meaning of stem s (see the $+$) and that of each of its orthographic neighbors r_x (all words containing stem)

$$OSC_s(s) = \frac{\sum_{x=1}^k \cos(s, r_x) * f_{r_x}}{\sum_{x=1}^k f_{r_x}}$$

Cluster-centered variation: Mean cosine similarity between meanings of orthographic neighbors and frequency-weighted cluster centroid (see the X)

$$OSC_c(s) = \mathbb{E} \left[\cos(r_x, \frac{\sum_{j=1}^n r_j * f_{r_j}}{\sum_{j=1}^n f_{r_j}}) \right]$$

Calculation details

Input values

- GloVe (Pennington et al., 2014) for semantic vectors
- CELEX (Baayen et al., 1995) for word frequencies; vocabularies of CELEX and SUBTLEX-US (Brysbaert and New, 2009) for orthographic neighbors.

Versions of OSC for metric assessment

- OSC_c - new cluster-centered variation
- OSC_s - stem-centered version calculated with GloVe and CELEX inputs
- OSC_m - original stem-centered values from Marelli and Amenta (2018)

Assessing cluster-centered variation with data from prior studies

Explaining unprimed lexical decision latencies

Compared how well three OSC metrics predicted lexical decision latencies for 8,857 words from English Lexicon Project (Balota et al., 2007). Word length, log frequency and log family size as covariates.

Model	OSC Effect	Variance Explained	AIC	BIC
Baseline		0.41	-30370.91	-30335.46
OSC_c	-0.0029	0.41	-30369.74	-30327.21
OSC_s	-0.015***	0.42	-30412.08	-30369.55
OSC_m	-0.011***	0.42	-30403.65	-30361.12

Explaining masked morphological priming magnitudes

Compared OSC metrics as predictors of masked morphological priming magnitudes for 676 prime-target pairs from prior lexical decision studies (Rastle et al., 2000, 2004; Andrews and Lo, 2013; Jared et al., 2017). Target length, log family size, orthographic neighborhood, transparency, and prime and target frequency as covariates, random effects for target and study.

Model	OSC Effect	Variance Explained	AIC	BIC
Baseline		0.20	4826.09	4867.83
OSC_c	43.89*	0.19	4814.16	4860.07
OSC_s	28.85*	0.18	4815.18	4861.09
OSC_m	23.22**	0.19	4813.23	4859.15

Summary of metric assessment

- Stem-centered OSC metric: better predictor for unprimed lexical decision latencies.
- Cluster-centered OSC metric: as good or better predictor of masked morphological priming magnitudes.
- Cluster-centered metric appears to be informative as morphological metric, and more flexible.

Acknowledgements

Many thanks to Professors Sally Andrews and Debra Jared for sharing their data for these analyses. The research reported here was supported in part by a training grant from the Institute of Education Sciences (R305B150008). Opinions expressed do not represent the views of the U.S. Department of Education.

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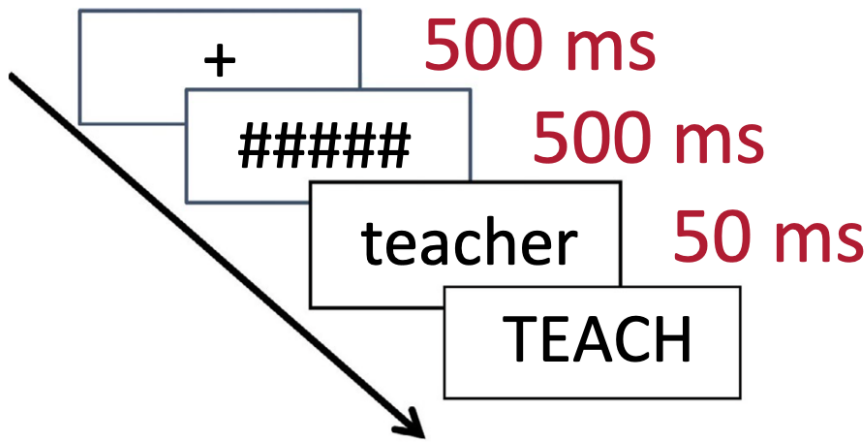
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Experiment Motivation

- TEACHER-teach = CORNER-corn > CASHEW-cash (Rastle et al., 2004)
- TEACHER-teach > CORNER-corn = CASHEW-cash (Jared et al., 2017)
- OSC is a common confound (Marelli et al., 2015)
- How does varying OSC_c impact these effects?

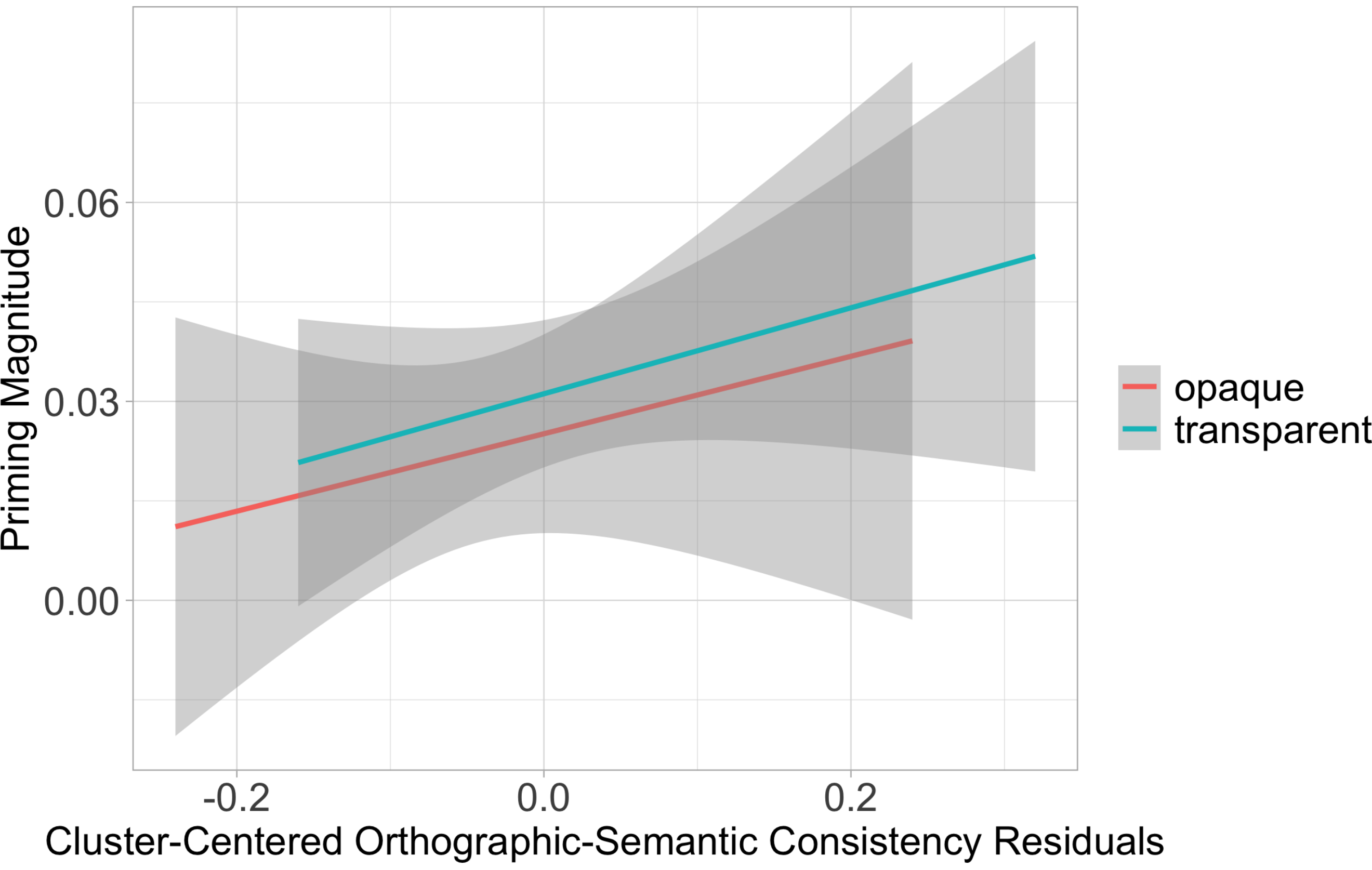
Method

- 100 words per condition (transparent and opaque)
- maximally varied OSC within each condition
- OSC minimally correlated with other factors (length, freq, family size)
- 44 undergraduates at a private university in southwestern Pennsylvania.



Results

- Analyzed separately: Interactions between OSC and prime relatedness not significant (opaque: $t = -0.10$, $p = 0.12$; transparent: $t = -0.051$, $p = 0.32$)
- Analyzed together: Marginal interaction of OSC and prime relatedness ($t = -0.073$, $p = 0.063$)



Conclusions

Cluster-centered orthographic-semantic consistency provides a more flexible metric for meaning carried by a particular letter string. Experimental results suggest both opaque and transparent morphological priming are stronger when the shared stem has higher OSC.